SR series
Industrial Robot

MAINTENANCE MANUAL

(SEPARATE VOLUME)

(SR–554HSP–T–Z–S)

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before installing and operating the industrial robot, read through and completely understand this manual.
   After reading through this manual, keep it nearby for future reference.

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NUMAZU, JAPAN

Shibaura Machine

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PREFACE

This manual constitutes a separate volume of the SR–554HSP industrial robot maintenance manual (STE 71371).
It describes the differences between the standard specifications.
Be sure to look through this manual and set up a maintenance program before actually unpacking the shipment.

PRECAUTIONS FOR SAFETY

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)."

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.

*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). The details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). The details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger and caution. The details of the actual danger and caution are indicated with pictures or words in or near the symbol.

CAUTION

- Be sure to read through the safety instruction manual provided separately before starting actual work to assure the safe work covering from the robot installation to operation.

[Maintenance and inspection]

Be sure to observe the following items to use the robot safely.

DANGER	
 Prohibited	• DO NOT incinerate, disassemble or charge the batteries. Otherwise, they may rupture.
 Mandatory	• Be sure to turn off the main power switch of the controller before starting inspection or maintenance. • Batteries should be disposed of according to the user's in-house regulations.

CAUTION	
 Disassembly prohibited	• The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.
 Mandatory	• Always use the SHIBAURA MACHINE's designated spare parts when replacing the parts. • Maintenance and inspection should be performed regularly. Otherwise, the system may malfunction or accidents will be caused.

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Section 1 Maintenance of the Robot

1.1 Cautions on Maintenance and Inspection

When performing inspection or maintenance of the robot, strictly observe the following precautions to protect yourself and coworkers.

DANGER

- Be sure to turn off the main power switch of the controller before approaching the robot for maintenance and inspection.

1.2 Layout of Robot Components and Drive Mechanism

The layout of the robot mechanical components is shown in Fig. 1.1.

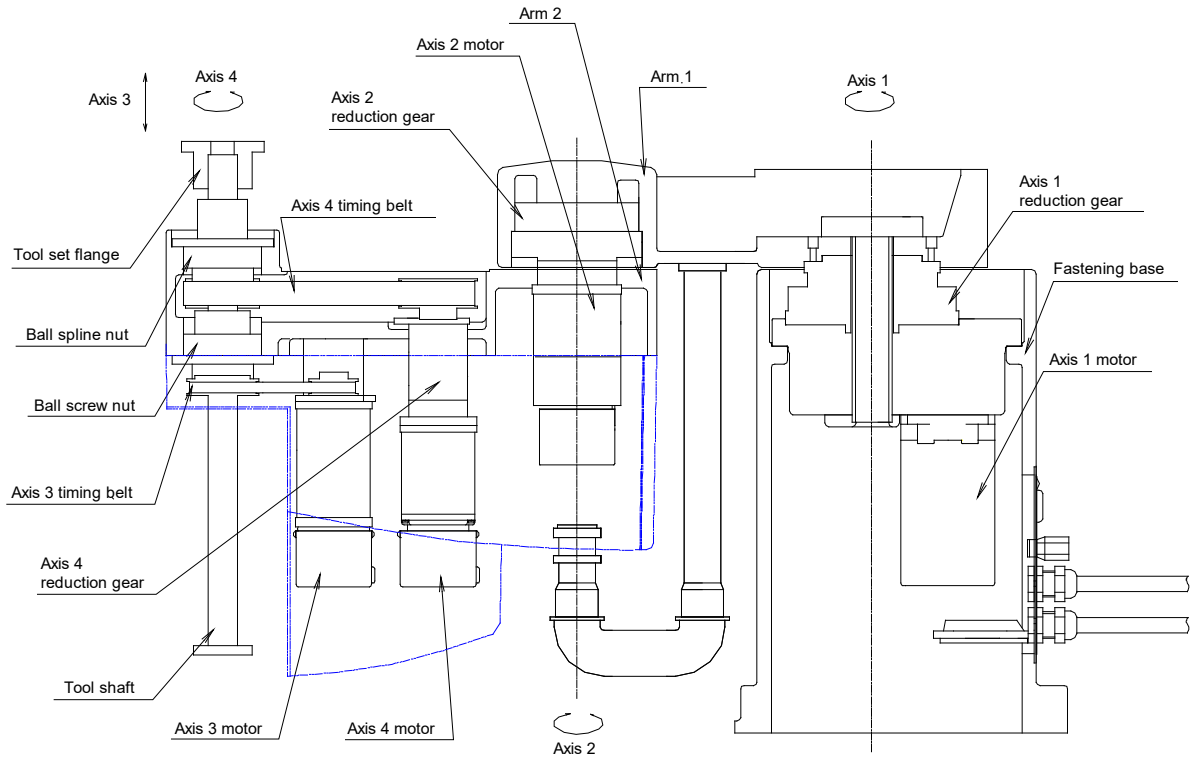


Fig. 1.1 Layout of robot mechanical components

The drive mechanism of each axis is described below.

a) Axis 1
The revolution of the axis 1 AC servo motor is input to the axis 1 reduction gear via the gear. The output of the axis 1 reduction gear directly controls the arm 1 motion.

b) Axis 2
The revolution of the axis 2 AC servo motor is input to the axis 2 reduction gear. The output of the axis 2 reduction gear directly controls the arm 2 motion.

c) Axis 3
The revolution of the axis 3 AC servo motor is input to the ball screw nut through the timing belt to move the tool shaft up and down.

d) Axis 4
The revolution of the axis 4 AC servo motor is input to the reduction gear. The output of the reduction gear is input to the ball spline nut through the timing belt to revolve the tool shaft.

1.3 Robot Maintenance Procedures

1.3.1 Check of Cable Duct for Abnormality

a) Visually check to see that the cable duct is not locally twisted, bent or cracked when the robot moves. If there is any abnormality, replace the cable to prevent breakage of the cable wire, etc.

b) Each connector securing the cable duct is secured by tightening it to the plate with a hexagon nut. If the hexagon nut is loosened, tighten it with a pair of water pump pliers, etc.

1.3.2 Check of Tool and Base for Mounting

! DANGER

- Completely tighten the tool clamp bolt. Otherwise, the tool may drop or shift from the predetermined position.

a) Check of tool set flange bolt for looseness
Make sure, using a hexagonal wrench key, that the hexagon socket head cap screw of the tool set flange that is tightened onto the tool shaft is not loosened. Otherwise, tighten it.
Also make sure by means of the hexagonal wrench key that the hexagon socket head cap screw securing the tool set flange and tool is not loosened. Otherwise, tighten it.

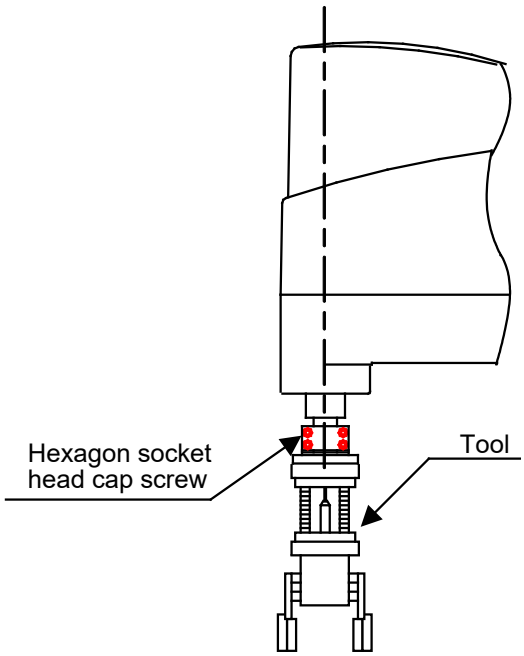


Fig. 1.2 Tool set area

b) Check of basic robot installation bolts
Make sure, using the hexagonal wrench key, that the installation bolts of the robot base are not loosened. Otherwise, tighten them.

1.3.3Check of Air Leakage

Piping connections in the robot are shown in Fig. 1.3. Make sure that no air leaks from the robot interior or air joints.

DANGER

- Be sure to turn off the main power switch of the controller before approaching the robot for maintenance and inspection.

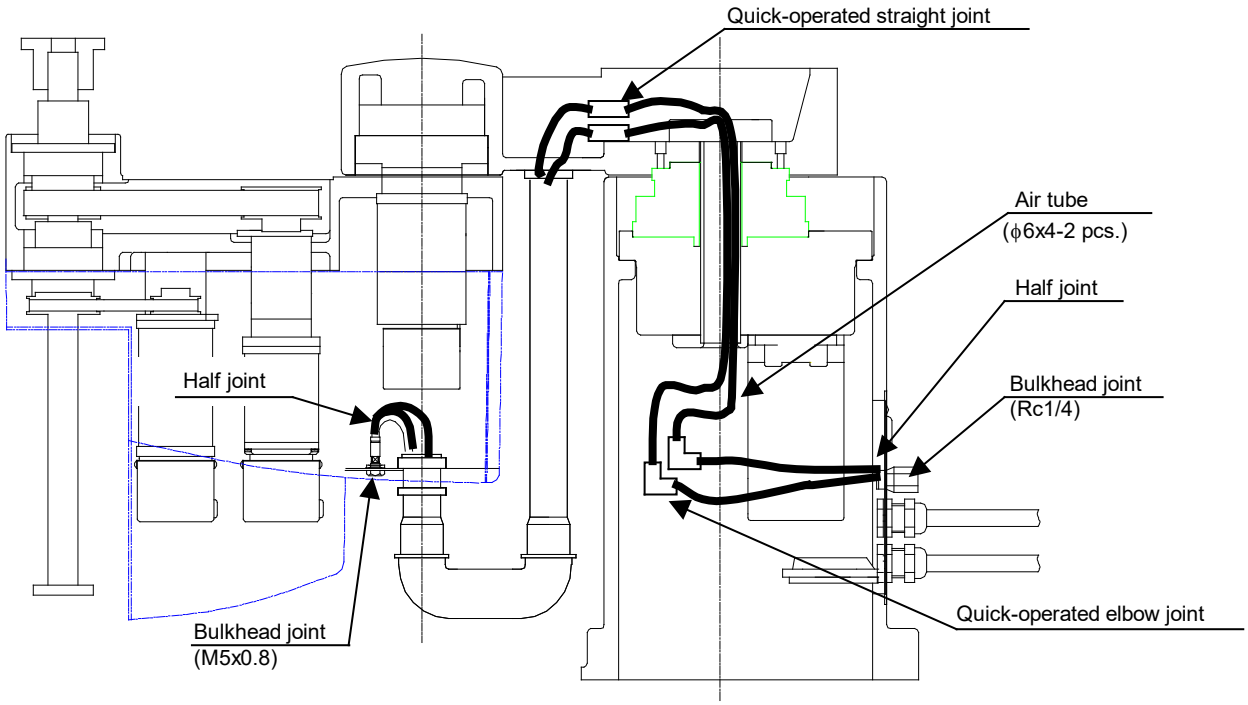


Fig. 1.3 Tool air piping

1.3.4 Lubrication of Tool Shaft

To lubricate the tool shaft, use the following grease.

- FOMBLIN grease OT–20 (maker: Solvay Solexis)

! CAUTION

- DO NOT mistake the type of grease. Otherwise, the robot performance may deteriorate or the robot may malfunction.
- After you have touched the tool shaft by hand, be sure to apply the grease to it. Otherwise, rust will develop on the shaft.

a) Check of tool shaft for lubrication
Open the arm 2 cover and make sure that the tool shaft has not run out of grease and that no rust has developed on it. If the tool shaft has run out of grease, apply the grease according to Para. b).

b) Greasing of tool shaft
Apply the grease with a brush or by hand so that all guide slots around the shaft are covered. Just after the greasing, DO NOT start an operation. First, move the tool shaft slowly over the entire motion range. Then wipe off any grease that oozes from the shaft.

c) Greasing of tool shaft support bearings
Fill the specified volume of grease from the grease nipples at the end of the arm 2, using a grease gun. The greasing volume is 5 cm^3, and the greasing locations are shown in Fig. 1.4.

Type of grease nipple: A–M6F

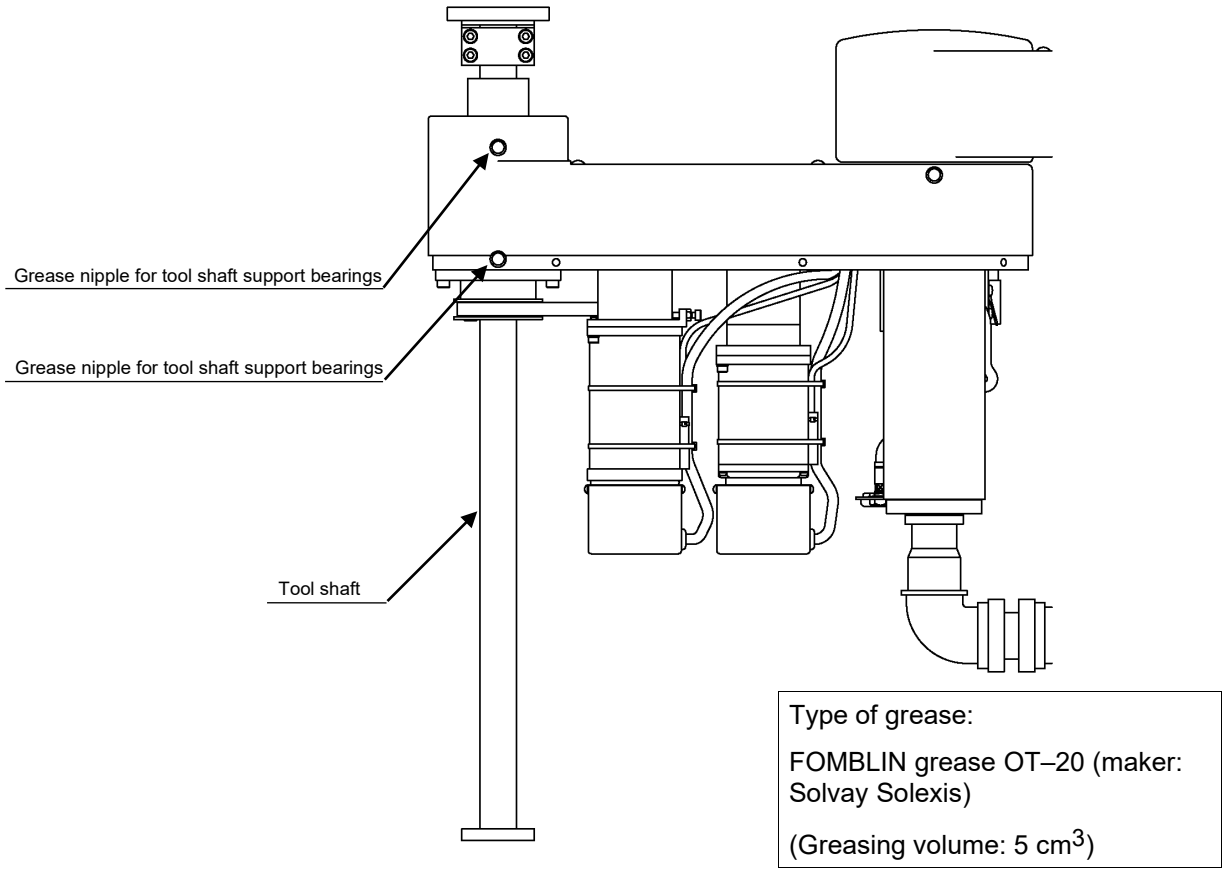


Fig. 1.4 Greasing locations of tool shaft support bearings

1.3.5 Check of Each Axis Drive Status

Move each axis by hand and make sure that it can move smoothly. Perform check on the axis 3 and axis 4 while they are in the servo-free status.

! DANGER

- When nearing the robot in the servo-free status, be sure to carry out the work by two (2) persons. One person should carry out the work while the other is monitoring the work outside the hazardous zone. The latter person should watch the work and be ready to press the emergency stop switch if any abnormality occurs.

1.3.6 Check of Timing Belt Status

Remove the arm 2 cover and arm 2 lower cover, and visually check to see that the belts of the axis 3 and axis 4 are not worn off or cracked.
Also check the timing belt for tension and adjust as necessary.

1.3.7 Greasing of Reduction Gears

Remove the covers and fill the grease from the grease nipples, using a grease gun. The greasing volume is 10 cm^3 (initial time 20 cm^3) for the axis 2. For the axis 1, fill an appropriate volume of grease while checking for the volume from the sight hole.
The greasing locations are shown in Fig. 1.5.
The type of grease nipple is A–M6F.
As the axis 4 reduction gear is of a non-greasing type, it need not be greased at all.

Type	Greasing location	Type of grease	Greasing volume
SR–554HSP–T	Axis 1	Alvania grease 0	Appropriate volume
	Axis 2	Harmonic grease 4B No.2	10 cm^3 (Initial time 20 cm^3)

! CAUTION

- DO NOT mistake the type of grease. Otherwise, the robot performance may deteriorate or the robot may malfunction.

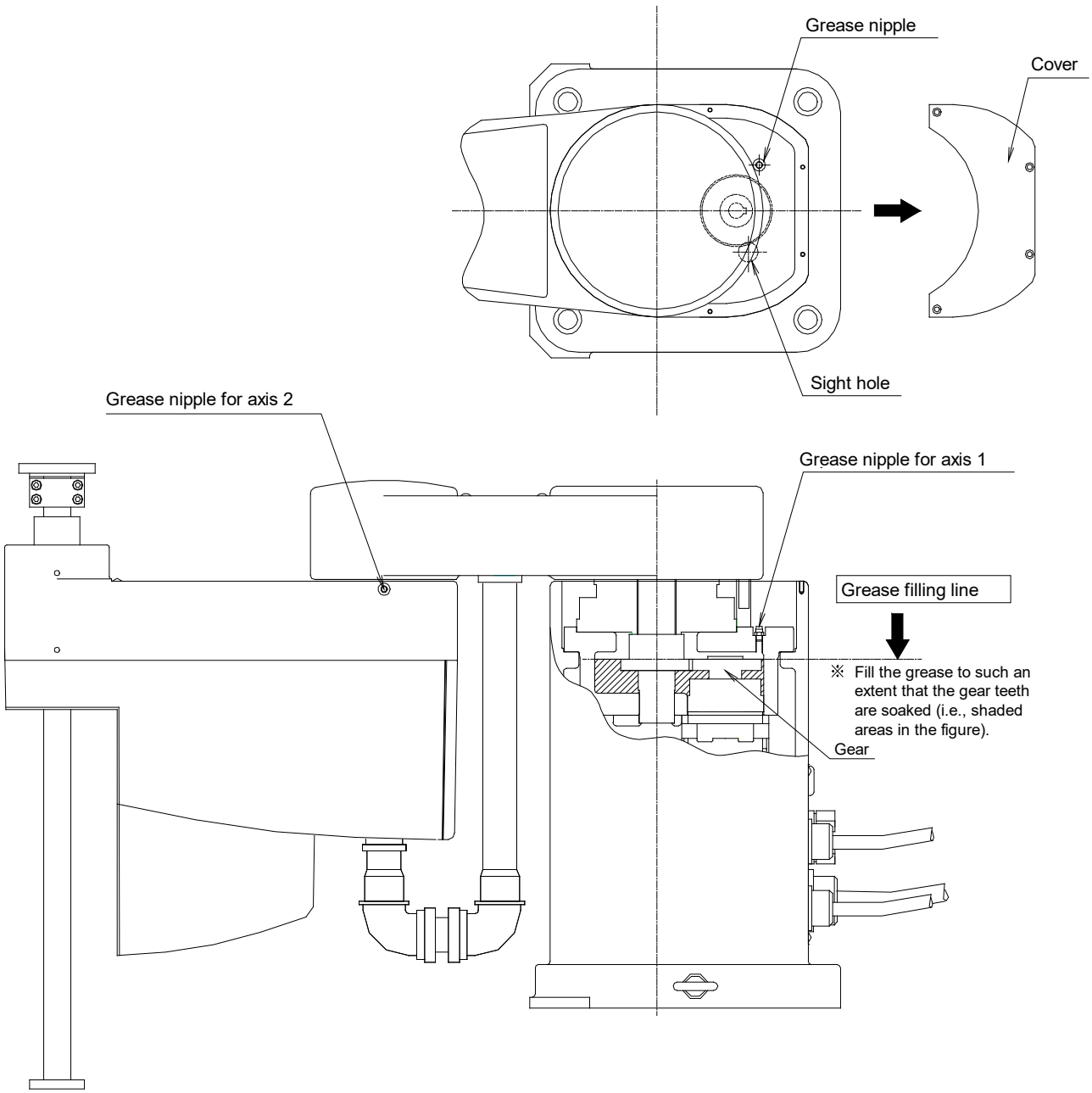


Fig. 1.5 Greasing locations of reduction gears

1.3.8 Replacement of Position Detector (or Encoder) Batteries

To back up the data of the position detector attached to each motor, batteries (lithium batteries) are used. Replace the batteries every three (3) years.
If the controller has detected the battery voltage drop alarm, however, replace the batteries at an early stage
The batteries should be replaced in the following manner according to the battery charged time.

[Replacing procedures]

1) Turn on the main power switch of the controller for more than one (1) minute.
2) Turn the power off.
3) Remove the cover shown in the figure below.
4) The lithium batteries are secured to the battery plate by means of a band. Remove them, using a nipper, etc.
   Remove the lithium batteries from the connector (BATT) and replace them with new ones within five (5) minutes.
5) Secure the new lithium batteries to the battery plate, using the band.
6) Mount the cover on the robot base.

Now, the replacement has finished.

! CAUTION

- Batteries should be disposed of according to the user's in-house regulations.
- NEVER drop the batteries into fire. NEVER short-circuit, charge, disassemble or heat them. Otherwise, liquid leakage or rupture may be caused.

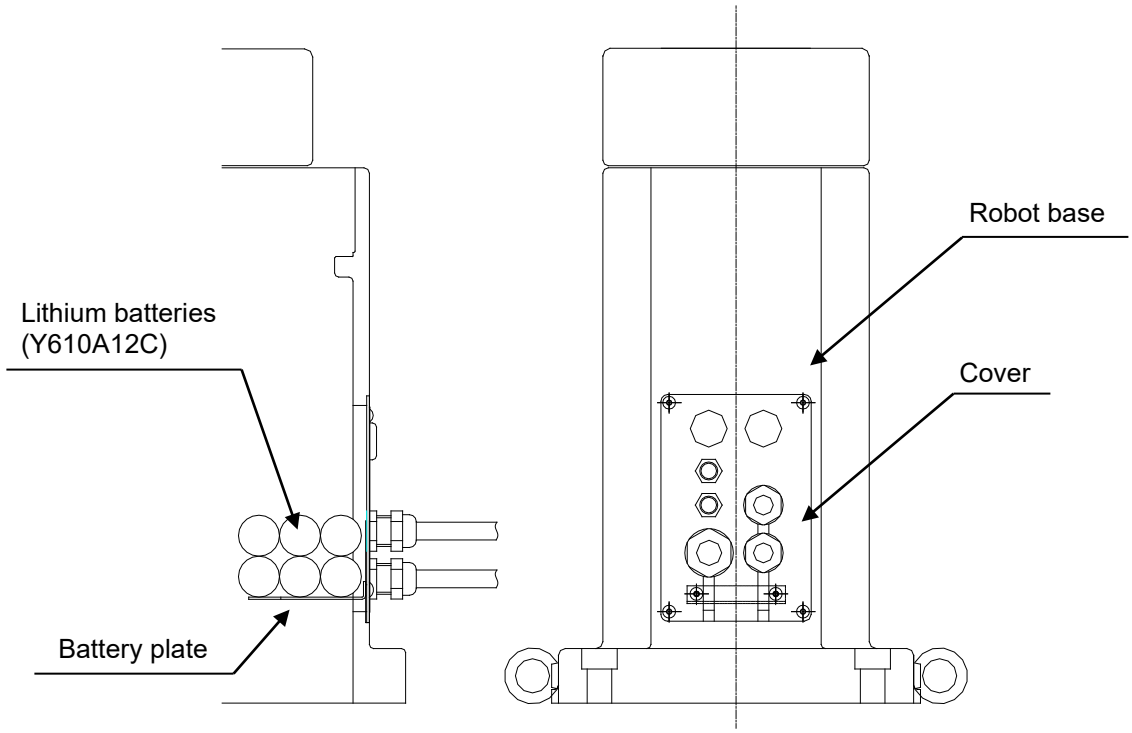


Fig. 1.6 Position detector battery set position

Section 2 Replacement Parts for Maintenance

The maintenance parts for the SR–554HSP–T–Z–S robot are given in the table below.

List of replacement parts for maintenance

Part name	Type	Maker	Q'ty	Remarks
AC servo motor	VLBST–Z06030S–OU001	SHIBAURA MACHINE	1	Axis 1
	VLBST–Y04030S–OU001		1	Axis 2
	VLBST–Z02030S–BOU002 VLBST–Z02030S–BOU005 *1)		1	Axis 3
	VLBST–Y01030S–BOU001 VLBST–Y01030S–BOU008 *1)		1	Axis 4
Reduction gear	773AD0207		1	Axis 1
	M0713566P1		1	Axis 2
	L890190 S816851 *2)		1	Axis 4
Timing belt	339–3GT–9		1	Axis 3
	565–5GT–20 S816852 *2)		1	Axis 4
Timing pulley	K0713127P1		1	Axis 3
	K0713126P1		1	Axis 3
	K0713128P1 S777592 *2)		1	Axis 4
	K0713125P1		1	Axis 4
Ball screw spline	S860351		1	Axis 3
Robot harness	E95163G01		1	
Base harness	E95164G01		1	
Cable	KH6665G01		1	
Grease	FOMBLIN grease OT–20	Solvay Solexis	100 g	
	Alvania grease 0	Shell Oil	2.5 kg	
	Harmonic grease 4B No.2	Harmonic Drive Systems	500g 2.5 kg	

・Please contact SHIBAURA MACHINE at order entry, after you confirm the serial number of the main robot.

*1) These motors are new type. They are used instead of the motors of the old type.

*2) These parts are the maintenance parts for the shipment machine after June, 2008. When you exchange the reduction gear (L890190) to the reduction gear (S816852), you need timing pulley (S777592), the plate (S777591) to mount the motor, six (6) hexagon socket head cap screws (M4X25), eight (8) hexagon socket head cap screws (M5X16) and eight (8) washer (GT-SH5). Also it is necessary to change the parameter of the controller.

! CAUTION

- With the exception of the grease, all parts listed above are custom-made to SHIBAURA MACHINE specifications. Contact SHIBAURA MACHINE at order entry.
- NEVER replace the parts by the user.